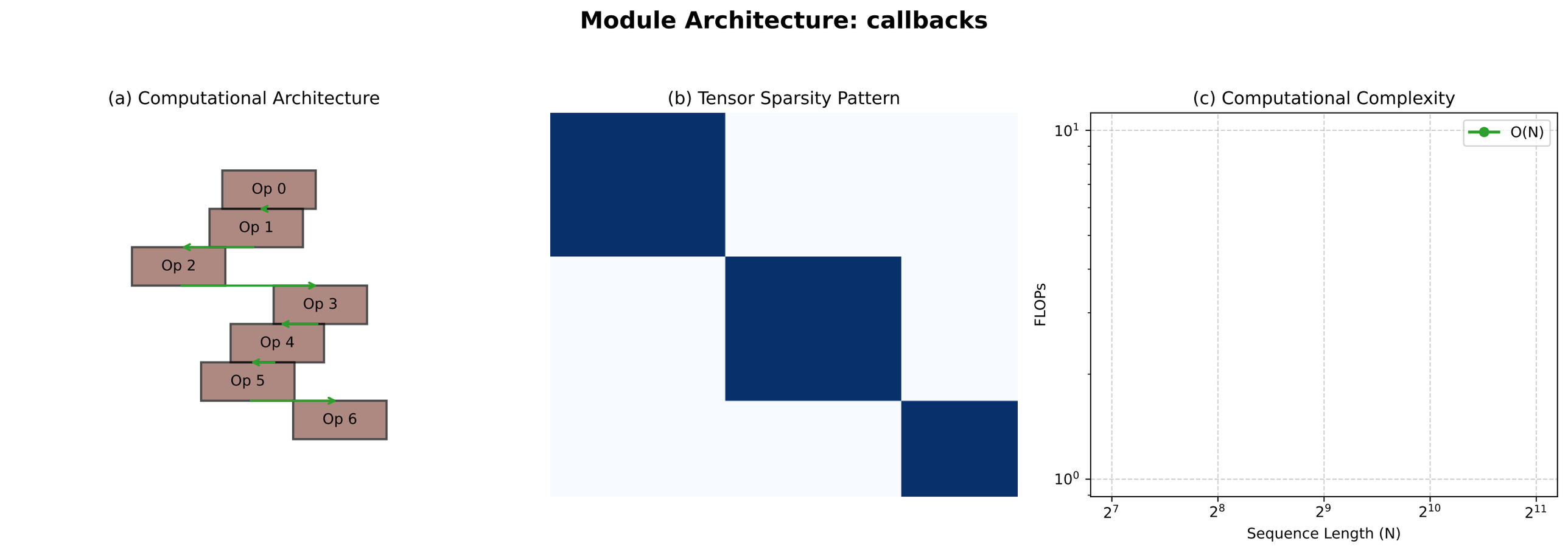Module Architecture: callbacks